

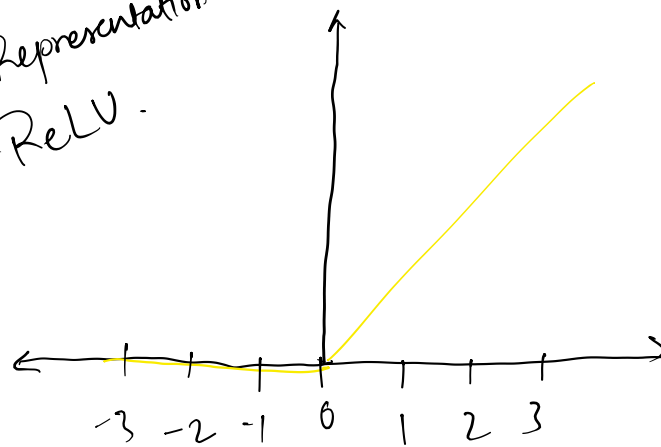
There are two types of activation functions:-

- ① Linear Activation function
- ② Non-Linear Activation functions

① Binary Activation :- A threshold is defined, and the values above the threshold would give the output 1. The values below the threshold would give output as 0.

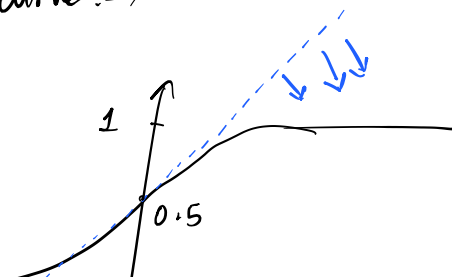
② ReLU Activation function :- The Rectified Linear Unit [ReLU] activation function shows 0 as the output for all negative inputs.

Graphical Representation for ReLU.



③ Sigmoid function → Sigmoid functions are used to converge the output to a specific range of 0 to 1.

The graph for sigmoid curve:-



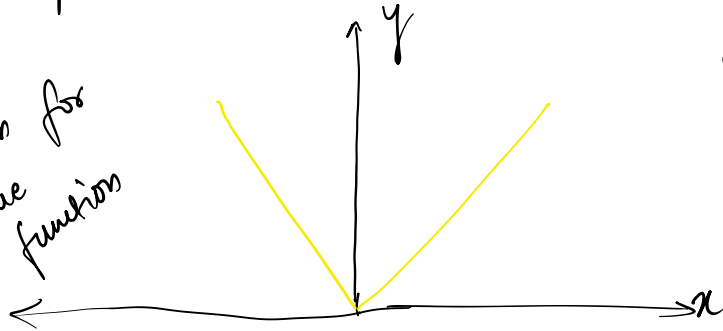
Mathematical formula

$$f(x) = \frac{1}{1 + e^{-x}}$$



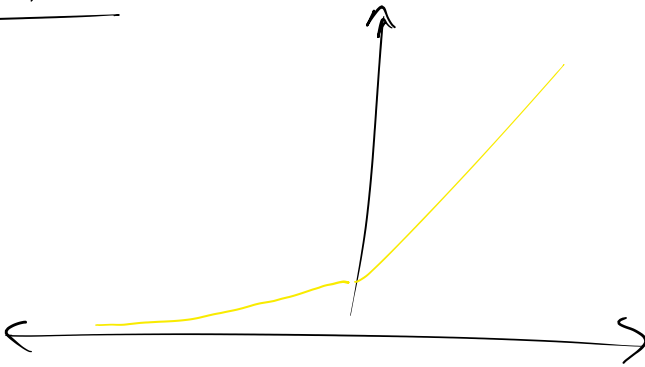
- ④ Absolute Value Rectified The Absolute value rectified activation function gives always a positive output.

Graphical Representation for Absolute value Rectified function



$$f(x) = |x|$$

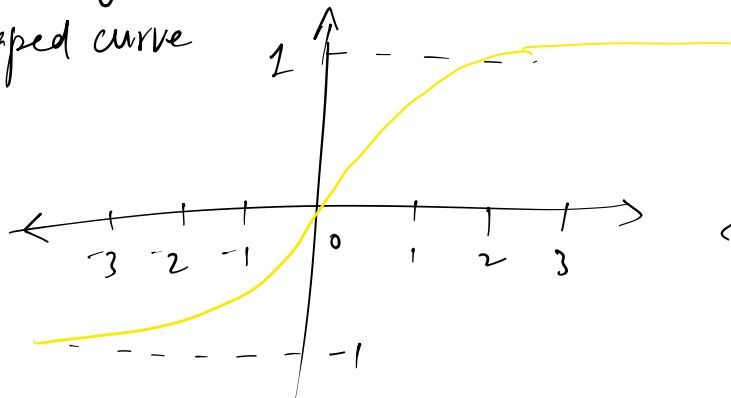
- ⑤ Leaky ReLU



- ⑥ Tanh Activation function : Hyperbolic Tangent [tanh]

converges everything to the range of $[-1, 1]$

Graph is S-shaped curve



Mathematical formula

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$